STATUTORY AUDIT TRAINING SUITE • FY 2025–26 (AY 2026–27)

The First-Year Audit Trail: User Manual & Technical Guide

Complete Operational Handbook & System Architecture Reference for Browser & Full-Stack Python Editions

1. System Overview & Deployment Editions

The First-Year Audit Trail is an interactive statutory audit training platform engineered for first-year Chartered Accountancy article trainees and junior audit executives. Using simulated client records for **Apex Trading Pvt Ltd** (Mumbai, GSTIN: `27AABCA1234F1Z5`), trainees execute real-world audit procedures across primary vouching, bank reconciliation, and Finance Act 2025 TDS scrutiny before generating an executive Working Paper Observation Schedule.

The platform ships in two complementary editions so audit teams can run zero-setup individual exercises or host multi-trainee firm-wide sessions over an office local area network (LAN).

Feature / Layer	Standalone Browser Edition (`index.html`)	Full-Stack Python Suite (`Audit_Trail_FullStack`)
Runtime Stack	Vanilla ES6+ JavaScript, Tailwind CSS, Web Audio API	Python 3.9+, NiceGUI, FastAPI, Pandas, OpenPyXL, SQLite3
Installation	Zero installation; double-click `index.html` in any browser	Automated `.audit_venv` setup & PyPI install via bootstrapper
Data Source	Embedded statutory scenarios for FY 2025–26	Randomized figure generator + real Tally `.xlsx` / `.csv` upload
Score Persistence	In-memory browser session state	Persistent SQLite LMS database (`audit_training_lms.db`)
Partner Supervision	Printable A4 Working Paper Memo (`window.print()`)	Live Partner Leaderboard, Rule Miss Analytics & Styled `.xlsx` Export

2. Installation & Automated Full-Stack Setup

The Full-Stack Python Suite is designed for zero-touch provisioning. You only need **Python 3.9 or newer** installed on the host desktop or laptop (with *"Add Python to PATH"* enabled on Windows) and an active internet connection on the first run so the bootstrapper can fetch required packages.

2.1 Launching the Suite (Three Options)

1. **Windows One-Click Batch Launcher:** Extract `Audit_Trail_FullStack_Suite.zip` into any folder and double-click `START_WINDOWS.bat`.
2. **macOS / Linux Shell Launcher:** Open a terminal in the extracted folder, make the script executable once (`chmod +x start_mac_linux.sh`), and run `./start_mac_linux.sh`.
3. **Single-File Self-Extractor:** Place `audit_trail_fullstack_installer.py` into any empty folder and execute `python audit_trail_fullstack_installer.py`. It unpacks the entire directory tree and triggers the bootstrapper automatically.

2.2 Automated Bootstrapper Sequence (`bootstrap_launcher.py`)

When started, `bootstrap_launcher.py` executes a three-stage provisioning pipeline without manual intervention:

```
[1/3] Creates isolated virtual environment at ./Audit_Trail_FullStack/.audit_venv
[2/3] Verifies & installs: nicegui>=2.5.0, pandas>=2.1.0, openpyxl>=3.1.2, python-multipart>=0.0.9
[3/3] Initializes SQLite LMS DB, generates sample Tally .xlsx, detects LAN IP, and binds port 8080
```

Once step 3 completes, the console prints both your **Local Desk URL** (`http://localhost:8080`) and your **Office Wi-Fi URL** (for example, `http://192.168.1.45:8080`). Any trainee connected to the same office Wi-Fi or Ethernet switch can open that LAN URL in their browser to sign in and complete their own tracked audit file simultaneously.

3. Step-by-Step Trainee Operational Manual

Upon opening the portal, trainees enter their full name and batch designation (*1st Year Article*, *2nd Year Article*, *3rd Year Article*, or *Audit Executive*) to open a tracked audit file. In the Standalone Browser Edition, trainees begin immediately on Working Paper `W/P A-1` and can use keyboard shortcuts (`V` to Verify, `Q` to Raise Query, and `N` to toggle the Senior's Field Notebook).

3.1 Level 1 — W/P A-1: Vouching & Primary Documentation

Trainees inspect five realistic purchase and expense vouchers rendered on the left desk stage. Hovering over highlighted fields in the Browser Edition reveals inspection cues on buyer GSTINs, payment modes, tax splits, and physical attachments. Trainees select the applicable statutory finding on the right console and click either **STAMP VERIFIED (✓)** or **RAISE QUERY (?)**.

Scenario Area	Embedded Audit Defect / Condition	Expected Action & Statutory Rule
Computers & Laptops	MacBook tax invoice billed in Director's personal name without Company GSTIN; ITC claimed	**RAISE QUERY:** Ineligible ITC under Sec 16(2) CGST Act & personal-use check
Repairs & Renovation	Single-day physical cash payment exceeding ₹10,000 (e.g., ₹14,500) to carpenter	**RAISE QUERY:** 100% disallowance under Sec 40A(3) of Income Tax Act
Office Supplies	Local Mumbai supplier (`27...`) billing Apex Mumbai (`27...`) with 18% IGST	**RAISE QUERY:** Wrong tax head on intra-state supply (Sec 8/12 IGST Act)
Postage & Courier	Valid B2B BlueDart invoice with Company GSTIN, 9% CGST + 9% SGST, and dispatch log	**STAMP VERIFIED:** Clean & compliant voucher (false-positive trap check)
Printing & Stationery	₹8,400 booked solely on an internal staff petty-cash slip; external vendor bill missing	**RAISE QUERY:** Insufficient external audit evidence under SA 500

3.2 Level 2 — W/P B-1: Bank Reconciliation Detective

Working Paper `W/P B-1` presents Apex Trading's **HDFC Bank Book (Ledger)** alongside the external **HDFC Bank Statement (March 2026)** in a two-step workflow:

1. **Step 1 — Cross-Ticking Routine Entries:** Click an entry in the Company Bank Book and click its corresponding entry in the Bank Statement. Matching pairs (NEFT, clearing cheque, UPI bill payment, RTGS receipt) highlight in green with a `✓✓` cross-tick mark.
2. **Step 2 — Resolving Year-End Unmatched Exceptions:** Two unmatched items remain as at 31-Mar-2026: (a) **₹1,180 Bank Charges + 18% GST** debited directly in the bank statement on 28-Mar-2026 (requires booking `Dr. Bank Charges ₹1,000, Dr. Input GST ₹180 → Cr. HDFC Bank ₹1,180`), and (b) **Cheque #441209 for ₹25,000** issued on 10-Oct-2025, which has crossed the 3-month RBI cheque validity window (stale cheque requiring reversal: `Dr. HDFC Bank ₹25,000 → Cr. Vendor Account ₹25,000`).

3.3 Level 3 — W/P C-1: TDS & Sec 40(a)(ia) Compliance Spotter

Trainees scrutinize four Tally-style expense ledgers against the **Finance Act 2025** withholding thresholds applicable for FY 2025–26. Where a default is detected, non-deduction triggers a **30% expenditure disallowance risk** under Section 40(a)(ia) of the Income Tax Act.

> **Crucial Finance Act 2025 Threshold Trap (Effective 1 April 2025):** Section 194I rent withholding shifted from ₹2,40,000 per annum to **exceeding ₹50,000 per month (or part of a month)**, while Section 194J professional/technical fee thresholds increased from ₹30,000 to **exceeding ₹50,000 per financial year**. Consequently, Bhiwandi godown rent of ₹35,000–₹45,000/month is *compliant* without TDS in FY 2025–26 even though the annual payout exceeds ₹2,40,000.

Ledger Account	FY 2025–26 Threshold & Rate	Audit Conclusion
Legal & Professional Fees	Sec 194J: > ₹50,000 p.a. (@ 10% Professional / 2% Technical)	**QUERY:** Retainer > ₹50,000 credited with ₹0 TDS; 30% disallowance u/s 40(a)(ia)
Main Office Rent (Nariman Pt)	Sec 194I: > ₹50,000 per month or part thereof (@ 10% Land/Bldg)	**QUERY:** Monthly rent ₹60k–₹70k exceeds ₹50,000/month without 10% TDS
Godown Rent (Bhiwandi)	Sec 194I: > ₹50,000 per month (replaces old ₹2.40L annual limit)	**VERIFY (TRAP):** Rent ≤ ₹50,000/month requires ₹0 TDS in FY 2025–26
AMC Contractor Charges	Sec 194C: > ₹30,000 single bill OR > ₹1,00,000 annual aggregate (@ 2% Co.)	**QUERY:** Single corporate invoice > ₹30,000 paid without 2% TDS

3.4 Level 4 — W/P D-1: Observation Schedule & Working Paper Export

Working Paper `W/P D-1` consolidates every voucher stamp and BRS adjustment into a formal **Client Statutory Audit Observation Schedule**. Trainees can click **Print A4 Memo** for physical sign-off or click **Download Styled Excel Working Paper (.xlsx)** in the Full-Stack Suite to generate a two-sheet workbook containing an Executive Summary bar chart and a formatted Observation Schedule with a blank Management Response column.

4. Custom Tally Daybook Ingestion & Partner Dashboard

Audit seniors can test trainees on custom client data by uploading any `.xlsx` or `.csv` file via the **Import Tally** control in the top bar (a template is downloadable anytime via the **Sample Tally .xlsx** button). Each uploaded row is automatically evaluated by `classify_row()` in `app/audit_engine.py`.

Column Name	Data Type / Example	Rule Engine Role
`Date` & `Voucher_No`	`2025-06-18`, `CSH-014`	Displayed on voucher header and indexed in SQLite audit trail
`Ledger_Head` & `Party_Name`	`Repairs & Maintenance`	Routes TDS items (`rent`, `legal`, `amc`) to `W/P C-1` vs `W/P A-1`
`Party_GSTIN`	`27AAECS9988K1Z1` or `URD`	Checks prefix `27` against IGST vs CGST+SGST tax columns
`Payment_Mode`	`CASH`, `NEFT`, `CHEQUE`	Flags `CASH` where `Base_Amount > 10000` under Sec 40A(3)
`Base_Amount`, `CGST`, `SGST`, `IGST`	Numeric INR values	Computes net payout, GST head validity, and 30% Sec 40(a)(ia) base
`TDS_Section` & `TDS_Deducted`	`194J`, `194I`, `194C`, `NA`; `0`	Evaluates Finance Act 2025 thresholds when `TDS_Deducted == 0`

Column Name	Data Type / Example	Rule Engine Role
`Supporting_Doc` & `Narration`	Text description	Detects personal-name B2C bills (Sec 16(2)) & missing vendor invoices (SA 500)

Clicking **Partner Dashboard** in the header opens live firm-wide telemetry sourced from SQLite, displaying total sessions, average firm accuracy, trainee rankings (*Partner's Star Junior* at ≥90% accuracy with ≥8 items; *Field Audit Ready* at ≥75% accuracy with ≥6 items), and a pass-rate breakdown showing which statutory provisions trainees miss most frequently.

5. Technical Architecture, Database Schema & API Reference

The Full-Stack Suite separates UI rendering, statutory rule classification, persistence, and Excel document generation into clean Python modules inside `Audit_Trail_FullStack/`.

File Path	Architectural Responsibility
`bootstrap_launcher.py`	Creates `.audit_venv`, runs `pip install`, discovers LAN socket IP, sets `PYTHONPATH`, launches server
`app/main.py`	NiceGUI reactive UI pages, Web Audio JS bridge, FastAPI REST routes, state management
`app/audit_engine.py`	`AuditVoucher` dataclass, INR formatter, `classify_row()` rule engine, randomized batch generator, BRS dataset
`app/database.py`	SQLite3 connection factory, schema initialization, atomic `ON CONFLICT DO UPDATE` decision upserts, analytics queries
`app/wp_exporter.py`	OpenPyXL workbook builder for sample Tally daybook and multi-sheet Working Paper export with embedded `BarChart`

5.1 SQLite3 Database Schema (`data/audit_training_lms.db`)

All trainee activity is persisted locally in two relational tables with automatic accuracy recalculation on every stamp:

```
CREATE TABLE trainee_sessions (
    session_id    INTEGER PRIMARY KEY AUTOINCREMENT,
    trainee_name  TEXT NOT NULL,
    article_year  TEXT NOT NULL DEFAULT '1st Year Article',
    started_at    TEXT NOT NULL,
    completed_at  TEXT,
    total_items   INTEGER DEFAULT 0,
    correct_items INTEGER DEFAULT 0,
    accuracy_pct  REAL DEFAULT 0.0,
    rank_badge    TEXT DEFAULT 'In Progress'
);

CREATE TABLE audit_decisions (
    id              INTEGER PRIMARY KEY AUTOINCREMENT,
    session_id      INTEGER NOT NULL REFERENCES trainee_sessions(session_id),
    wp_module       TEXT NOT NULL,     -- 'A-1', 'B-1', or 'C-1'
    item_ref        TEXT NOT NULL,     -- Voucher ID or 'BRS-EX1' / 'BRS-EX2'
    ledger_head     TEXT NOT NULL,
    amount_inr      REAL NOT NULL,
    trainee_action  TEXT NOT NULL,     -- 'VERIFY' or 'QUERY'
    rule_selected   TEXT NOT NULL,
    expected_rule   TEXT NOT NULL,
    is_correct      INTEGER NOT NULL,  -- 1 (Accurate) or 0 (Review Needed)
    statutory_note  TEXT NOT NULL,
    recorded_at     TEXT NOT NULL,
    UNIQUE(session_id, wp_module, item_ref)
);
```

5.2 REST API Endpoints & Synthesized Audio Engine

- **`GET /api/leaderboard`:** Returns JSON containing the top 100 trainee sessions and aggregated rule-level pass rates for integration with internal firm dashboards.
- **`POST /api/scan-tally-excel`:** Accepts a multipart `.xlsx` or `.csv` file upload and returns instant JSON statutory classifications, expected query counts, and coaching explanations. Interactive OpenAPI documentation is served automatically at `http://localhost:8080/docs`.
- **Procedural Web Audio Engine:** Both editions synthesize tactile desk sound effects directly via browser `AudioContext` oscillators (135 Hz → 42 Hz triangle wave for rubber-stamp thuds, 880 Hz → 1320 Hz sine sweep for pen ticks, and C5–E5–G5 arpeggios for completion chimes), eliminating external MP3 dependencies and offline broken links.

5.3 Operational Troubleshooting

Symptom / Issue	Root Cause	Resolution
`Python was not found` on Windows	Python 3.9+ not installed or omitted from `PATH`	Install Python from python.org and tick *"Add Python to PATH"* during setup

Symptom / Issue	Root Cause	Resolution
Port `8080` already in use	Another local service or prior simulator instance is running	Close the existing terminal window or change `port=8080` in `run_server()` inside `app/main.py`
Trainees on Wi-Fi cannot open LAN URL	Host OS firewall blocking inbound TCP port `8080` on Public profile	Set office Wi-Fi network profile to *Private* or allow inbound TCP port `8080` in Windows Defender Firewall
No stamp sound heard on first click	Browser autoplay policy awaiting user gesture or audio muted	Ensure system volume is active and click the speaker toggle in the top HUD bar